

Section 1.1:

Hardware, Software and Computer Types

This section will take a look at hardware/software and the different types of computers.

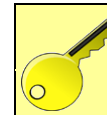
You will learn the differences between hardware and software, as well as examples of each.

You will also see different types of computers including their uses, advantages and disadvantages.

Key objectives of this section:

- | | |
|---|--|
| # | Know the difference between hardware and software and be able to identify examples of each . |
| # | Understand the uses of different types of computers and be able to identify examples of each . |
| # | Be able to discuss advantages and disadvantages of each different type of computer. |

Hardware



Key Words:

Hardware, Physical, See,
Touch, Pick-up

Definition: ***"All the physical parts of a computer that you you can see, touch and pick up."***

Overview of hardware:

Hardware is '**physical**'.

Physical means that it can be **touched** and **picked up**

Hardware **works with computer code** to carry out jobs such as:

- **Producing an image on a computer screen**
 - **Producing sound**
 - **Moving cursers / pointers around a screen**
 - **etc**
-

Examples of hardware: (click images to zoom and retrieve more info)

Keyboard



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Mouse



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Monitor



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Scanner



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Printer



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Other examples
of hardware
include:

- Graphics Card
- Sound Card
- Network Card
- Computer Case
- Cooling Fan
- Modem
- Router
- CD/DVD Reader
- CD/DVD Burner
- Power Supply
- Cables

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CD-ROM



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Motherboard



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Digital Camera



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Speakers



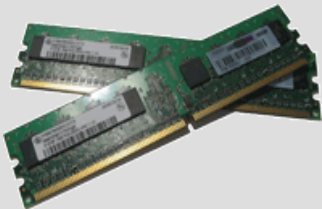
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Web Cam



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RAM



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Hard Disk



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Processor



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Microphone



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Memory Stick



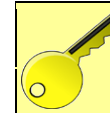
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NOTE:

We will look at
individual
computer
hardware
components in
a later unit.

Software

Definition: ***"Coded programs that control how the computer works"***



Key Words:

Software, Electronic,
Instructions, Cannot Touch or
Pick-up

Software **cannot be touched** or **picked up**.

This is because software is **not physical**. It exists as computer code inside the computer.

There are **2 types** of software - **System Software** and **Application Software**.

It refers to the **programs** that we run on our computer and it **tells the hardware** how to do something **useful**.

Software is basically a **set of instructions** (written in computer code) that tells the hardware what to do

Without software, most hardware would be **useless**.

Examples of software:

System software:

This type of software includes operating systems and different utilities to allow your system to run smoothly. System software often manages the computer's resources.

Operating Systems



Disk Cleaners



Disk Defragmenter



Virus Protection



Other examples of system software:

- Graphic Drivers
- Debuggers to identify system errors
- Compression Software
- Encryption Software
- Disk Check

Application software:

Everyday programs that you use to produce something useful. Microsoft Word and PowerPoint are good examples.

Word Processors



Graphics Editors



Web Design Programs



Spreadsheet Programs



Other examples of application software:

- Presentation Program (PowerPoint)
- Database Program (Access)
- Web Browser (Internet Explorer)
- CAD Design Software
- Media players
- Computer Games

Revision tips!

How to remember the difference between hardware and software.

1. Is it classed as a computer or part of an ICT system? **YES**
2. Can you touch it? **YES**
Then it is **HARDWARE**

1. Is it classed as part of an ICT system? **YES**
2. Can you touch it? **NO**
Then it is **SOFTWARE**

NOTE: If you are asked to name types of software you should **NOT** use brand names.

For example: you should use '**Graphics Editor**' rather than '**PhotoShop**'.

Activities!

[Hardware/Software - Task 1](#)

Click the above task and correctly identify the different examples of hardware and software.

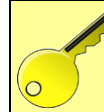
[Hardware/Software - Task 2](#)

Click the above task and write down as many examples of hardware/software you can think of.

Types of Computers

Definition of a Computer:

"Computers are devices that accept information and then manipulate it to produce a result."



Key Words:

Personal Computer,
Laptop, PDA, Mainframe,
Notebook, Palmtop

Computers allow us to perform many different tasks from gaming to complicated calculations. Computers range from very large to very small and it is estimated that there are over 1 billion of them in use around the world.

There are many different types of computers and people often choose which type they use based on what sort of tasks they want the computer to carry out. This section covers the main types of computers.

1 Personal Computer (Desktop PC)

- # Personal Computers are usually **found in the home**.
- # They are designed to be used by **one person at a time** and perform tasks such as **school work, gaming, household accounts, internet access** and **emailing**.
- # Personal Computers are made up of **input devices, output devices** and **storage devices** all held together within a **base unit** which houses all the hardware components that make the PC work.

Advantages (compared to laptop)

You get a **better specification** for your money.

(better components for the same price)

Disadvantages (compared to laptop)

Desktop PC's are **large** and **bulky**.

(this makes them not very portable)

Examples:



Cheaper to buy due to components being easier to make.

They need a **constant power supply** as they do not use batteries.

The **keyboard is easier to use** as it is larger than a laptop keyboard.

Need to copy files onto portable storage (memory stick) or send via email if you need to **work in a different location**.

Design of casing allows for **heat to escape**.

This helps to keep the PC pretty cool.

Note: This is because they are not portable and can't be moved easily.

Desktop PC

Output Device
(Monitor)

Base Unit
(Contains fragile electronics)

Input Device
(Mouse)

Input Device
(Keyboard)

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2 Laptop

- # Laptops are **portable computers** and are small enough to sit on your lap.
- # They can be **used almost anywhere** due to their **small size** and a **built-in battery** which powers the laptop when there is no access to a plug socket.
- # Modern laptops are generally **almost as powerful as a Desktop PC** and are used for similar tasks and purposes.
- # They have **LCD screens** (Liquid Crystal Display) which use **low-power**. This helps increase the run-time of a laptop's battery.

Examples:

Advantages (compared to PC's)

Disadvantages (compared to PC's)

Smaller and lighter and are much easier to transport. (portable)	Limited battery life means you need to recharge quite frequently.
Laptop components use low-power .	Smaller keyboards are sometimes hard to use.
Built-in rechargeable battery means they will work even without a mains power supply . (portable)	Laptops are prone to theft as they are often used in public places.
Usually come with built-in Wi-Fi which means they can access the internet from locations with access.	Laptops are compact and can get hotter than a desktop PC.
No need to transfer files onto portable storage if you need to work in a different location. You simply take the laptop with you.	



Laptop



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3 Mainframes

Mainframes are **very large and expensive** and are capable of supporting hundreds, or even thousands, of users at the **same time**.

Examples:

- # They are used in **large organisations** for **processing huge amounts of data** or to **control massive networks** of computers.
- # Mainframes use **more than one processor** (parallel processing) which means they can process **many jobs at the same time**.
- # Early mainframes were the **size of a room** and got their name because they were housed in **large metal 'frames'**.

Advantages (compared to other computers)	Disadvantages (compared to other computers)
They can process huge amounts of data very quickly.	Very expensive to buy (and to run).
Can perform millions and millions of calculations per second.	They are very large and so are almost impossible to transport .
Parallel processing means they can handle many tasks at the same time.	They generate a lot of heat and need to be in air-conditioned rooms.
They can run and manage huge networks of computers.	They need a high-level of training for people to be able to use them.



4 Netbook

- # Netbooks are **extremely small and light** versions of laptops and typically **weigh about 6 pounds**.
- # Netbooks can be used for **similar tasks to a laptop** but they are **not as powerful** and have **much smaller screens**.

Examples:

Advantages (compared to laptop)	Disadvantages (compared to laptop)
Much smaller and lighter... means they are even easier to transport.	Very small keyboards can be difficult to use.
Longer battery life (their low-power components require less energy).	Low performance means they are not suitable for power-hungry tasks like video editing.
Cheaper because they do not come with certain components (such as CD/DVD drive).	Small screen make images and text difficult to see.
	Very small size makes them easy to misplace or get stolen .
	Small hard disk drives means that they can store less data .



Netbook



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5 PDA (personal digital assistant)

PDA's are handheld computers with facilities including:

- **Personal organiser** (birthdays, appointments etc)
- **Internet browsing**
- **Telephone calls**
- **Multimedia** (video, music etc)
- **etc**

Generally speaking, there is very little difference between a PDA and a modern mobile phone.

Examples:

Advantages (compared to other computers)	Disadvantages (compared to other computers)
Smaller and lighter... means they are very portable. (they literally fit in your pocket)	Very small keyboards can be difficult to use.
Easy to use anywhere... even standing on a train.	Some use a stylus pen instead of a keyboard. (some people find these tricky to use)
Can be used like a mobile phone to make calls.	Small screen make images and text difficult to see. Very small size makes them easy to misplace or get stolen. Limited power... especially compared to desktop PC's.



Personal Digital Assistant (PDA)



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6 Palmtop

Palmtop's are very **similar in computing power to PDA's** and are used for basic functions such as:

- **Calendars**
- **Phone number and address book lists**
- **Internet access**
- **Email**

Examples:

Palmtop's get their name simply because they are **designed to sit in the palm of your hand**. They often come with a small keyboard.

Advantages (compared to other computers)	Disadvantages (compared to other computers)
Smaller and lighter... means they are very portable. (they literally fit in your pocket)	Very small keyboards can be difficult to use.
Easy to use anywhere... even standing on a train.	Less memory means they cannot run powerful programs (such as graphic editors)
Some have built in cameras and microphones . (useful for applications such as Skype)	Small screen make images and text difficult to see. Very small size makes them easy to misplace or get stolen . Limited power... especially compared to desktop PC's.



Palmtop



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Activity!

[Types of Computers - Task](#)

Click the above task and follow the instructions given on the sheet. Make sure that you complete both parts of the task sheet.

